

Hypothesis

If we say that SIMPLETOTAL increments the total exactly $1 + \sum_{k=5n}^{7n} k$ times for some input of n , we can conclude the Big-O time complexity is $O(n^2)$ if, for a given input n and a positive constant c ,

$$1 + \sum_{k=5n}^{7n} k \leq c \cdot n^2.$$

The summation can be written as

$$1 + \sum_{k=5n}^{7n} k = 1 + \sum_{k=1}^{7n} k - \sum_{k=1}^{5n} k = 1 + \frac{7n(7n+1)}{2} - \frac{5n(5n+1)}{2} \Rightarrow 12n^2 + 6n + 1.$$

Similarly, we can write the following inequality substituting the summation with our polynomial:

$$12n^2 + 6n + 1 \leq c \cdot n^2.$$

Base case

In order to prove this for a growing input n , we first need a base case. We choose $n_0 = 1$ as a positive non-zero input.

$$12(1) + 6(1) + 1 \leq c \cdot (1)^2 \implies 19 \leq c.$$

The inequality is true where $c \geq 19$. Now we show that this relation holds for a growing input, say $n = x + 1$.

Inductive step

$$\begin{aligned} 12(x+1)^2 + 6(x+1) + 1 &\leq c \cdot (x+1)^2 \\ 12(x^2 + 2x + 1) + 6x + 7 &\leq c(x^2 + 2x + 1) \\ 12x^2 + 24x + 12 + 6x + 7 &\leq c(x^2 + 2x + 1) \\ 12x^2 + 30x + 19 &\leq c(x^2 + 2x + 1) \end{aligned}$$

Where $c = 19$, we prove that SIMPLETOTAL is bound by $O(n^2)$:

$$12x^2 \leq 19x^2.$$